Drift-Flux Modeling of Two-Phase Flow in Wellbores

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Summary

Drift-flux modeling techniques are commonly used to represent two- and three-phase flow in pipes and wellbores. Unlike mechanistic models, drift-flux models are continuous, differentiable, and relatively fast to compute, so they are well suited for use in wellbore flow models within reservoir simulators. Drift-flux models require a number of empirical parameters. Most of the parameters used in current simulators were determined from experiments in small-diameter (2 in. or less) pipes. These parameters may not be directly applicable to flow in wellbores or surface facilities, however, because the flow mechanisms in small pipes can differ qualitatively from those in large pipes. In order to evaluate and extend current drift-flux models, an extensive experimental program was initiated. The experiments entailed measurement of water/gas, oil/water, and oil/water/gas flows in a 15-cm-diameter, 11-m-long plexiglass pipe at eight deviations ranging from vertical to slightly downward. In this paper, these experimental data are used to determine drift-flux parameters for steady-state two-phase flows of water/gas and oil/water in large-diameter pipes at inclinations ranging from vertical to near-horizontal. The parameters are determined using an optimization technique that minimizes the difference between experimental and model predictions for phase in-situ volume fraction. It is shown that the optimized parameters provide considerably better agreement with the experimental data than do the existing default parameters.

Introduction

Multiphase flow effects in wellbores and pipes can have a strong impact on the performance of reservoirs and surface facilities. In the case of horizontal or multilateral wells, for example, pressure losses in the well can lead to a loss of production at the toe or overproduction at the heel. In order to model and thereby optimize the performance of wells or reservoirs coupled to surface facilities, accurate multiphase pipeflow models must be incorporated into reservoir simulators.

Within the context of petroleum engineering, the three types of pipeflow models most commonly used are empirical correlations, homogeneous models, and mechanistic models. Empirical correlations are based on the curve fitting of experimental data, and their applicability is generally limited to the range of variables explored in the experiments. These correlations can be either specific for each flow pattern or can be flow-pattern independent. Homogeneous models assume that the fluid properties can be represented by mixture properties, and single-phase flow techniques can be applied to the mixture. These models can also allow slip between the phases, and this requires a number of empirical parameters. Homogeneous models with slip are called drift-flux models.

Mechanistic models are, in general, the most accurate because they introduce models based on the detailed physics of each of the different flow patterns. From a reservoir simulation perspective, however, mechanistic models can cause difficulties because they may display discontinuities in pressure drop and holdup at some flow-pattern transitions. Such discontinuities can give rise to convergence problems within the simulator. One approach to avoid these convergence issues is to introduce smoothing at transitions. An alternative approach is to apply a homogeneous pipeflow model. The drift-flux model is in fact a simple mechanistic model for intermittent flows, and it is used within general mechanistic models when the flow pattern is predicted to be bubble or slug.

Homogeneous models have the advantages of being relatively simple, continuous, and differentiable. As a result, they are well suited for use in reservoir simulators. The simplest homogeneous models, which neglect slip between the fluid phases (i.e., the fluid phases all move at the same velocity), are not appropriate for use in reservoir simulators because they fail to capture the complex relationship between the in-situ volume fraction and the input volume fraction. Drift-flux models, by contrast, are a good choice for use in reservoir simulators because they do account for the slip between the fluid phases. Drift-flux models are additionally capable of modeling countercurrent flow,[1] which allows the heavy and light phases to move in opposite directions when the overall flow velocity is small or when the well is shut in. For this reason, the drift-flux model is used in a number of reservoir simulators [e.g., in the multi-segment well model in the ECLIPSE® (registered trademark of Schlumberger) black-oil and compositional reservoir simulators[2,3]].

Drift-flux models require a number of empirical parameters. Most of the parameters used in current simulators were determined from experiments in small-diameter (2 in. or less) vertical pipes. These parameters may not be directly applicable to wellbore flows, however, because the flow mechanisms in small pipes can differ qualitatively from those in large pipes (Jepson and Taylor[4]). It is therefore important that the drift-flux parameters be determined for wellbores and pipes of sizes of practical interest.

The overall goal of our work in this area is the development of optimized drift-flux models for use in reservoir simulators. In order to accomplish this, extensive experimental and modeling efforts were initiated. The experimental work, reported in detail by Oddie *et al.,*[5] entailed large-diameter (6 in.) pipeflow experiments performed at a variety of phase flow rates and pipe inclinations. In our current study, these unique experimental data will be used to assess existing drift-flux models and to determine drift-flux parameters for steady-state two-phase flows of water/gas and oil/water in large-diameter pipes or wellbores. The parameters are determined using an optimization technique that minimizes the difference between experimental and model predictions. It is shown that existing default parameters are suboptimal in many cases. Models to accurately represent the effect of pipe inclination are also presented.

The basic drift-flux model was first proposed by Zuber and Findlay.[6] It has since been refined by many researchers (e.g., Ishii,[7] Nassos and Bankoff, [8] Wallis,[9] Hasan and Kabir,[10–12] and Ansari *et al.*[13]) and has been widely used for modeling both liquid/gas and oil/water pipeflow (Flores *et al.,*[14] Petalas and Aziz[15]). The drift-flux model applies two basic parameters: the profile parameter C_0 and drift velocity V_d. Using these parameters (which in turn depend on the system variables), in-situ phase volume fractions (holdup) can be calculated from the phase flow rates.

This paper proceeds as follows. We first present the detailed drift-flux model used in this work. We then provide an overview

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of the experimental program and describe the type and quality of the data collected. Next, we compute drift-flux model parameters from these data using an optimization technique that minimizes the difference between the experimental data and model predictions for steady-state water/gas and oil/water flows. The effect of pipe inclination is also modeled.

Drift-Flux Model

The drift-flux model for two-phase flow[6] describes the slip between gas and liquid as a combination of two mechanisms. One mechanism results from the nonuniform profiles of velocity and phase distribution over the pipe cross section (see **Fig. 1**). The gas concentration in vertical gas/liquid flow tends to be highest in the center of the pipe, where the local mixture velocity is also fastest. Thus, when integrated across the area of the pipe, the average velocity of the gas tends to be greater than that of the liquid. The other mechanism results from the tendency of gas to rise vertically through the liquid due to buoyancy.

A formulation that combines the two mechanisms is[6]

$$V_g = C_0 V_m + V_d. \quad (1)$$

Here, V_g is the flow velocity of the gas phase, averaged across the pipe area; C_0 is the profile parameter (or distribution coefficient), which describes the effect of the velocity and concentration profiles; V_m is the volumetric flux (or average velocity) of the mixture; and V_d is the drift velocity of the gas, describing the buoyancy effect.

The average mixture velocity is the sum of the gas and liquid superficial velocities,

$$V_m = V_{sg} + V_{sl} = \alpha_g V_g + (1-\alpha_g)V_l, \quad (2)$$

where α_g is the in-situ gas volume fraction, averaged across the area of the pipe. The average flow velocity of the liquid phase is thus

$$V_l = \frac{1-\alpha_g C_0}{1-\alpha_g} V_m - \frac{\alpha_g}{1-\alpha_g} V_d. \quad (3)$$

For efficient application in a well model of a reservoir simulator, we require expressions for C_0 and V_d that are relatively simple to compute, continuous, and differentiable. As will be evident later in this paper, some of the characteristics of the different flow patterns can be captured through these parameters.

Our discussion of the drift-flux implementation closely follows the description given in *Eclipse Technical Description Manual*.[3] We include a full discussion here for completeness.

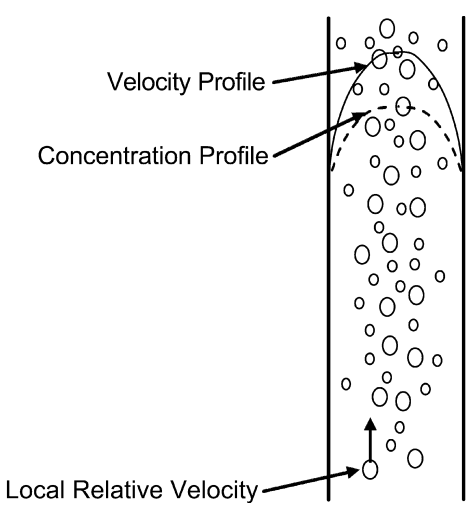

Fig. 1—Profile and local slip mechanisms in the drift-flux model.

The Profile Parameter. Zuber and Findlay[6] reported values of C_0 ranging between 1.0 and 1.5. Several drift-flux models use a value for C_0 of 1.2 in the bubble and slug flow regimes (e.g., Aziz *et al.*,[16] Ansari *et al.*,[13] and Hasan and Kabir[10]), but in the annular mist regime the value is close to 1.0. Moreover, C_0 should approach 1.0 as α_g approaches 1.0; in fact, $\alpha_g C_0$ should never exceed 1.0. Accordingly, we apply a relationship for C_0 that has a constant value at conditions equivalent to bubble or slug flow, and reduces to 1.0 as α_g approaches 1.0 or as the mixture velocity increases. A suitable expression with these properties is

$$C_0 = \frac{A}{1+(A-1)\gamma^2}. \quad (4)$$

The purpose of the term involving γ is to cause C_0 to reduce to 1.0 at high values of α_g or V_m. The γ parameter is given by

$$\gamma = \frac{\beta - B}{1-B} \quad \text{subject to the limits} \quad 0 \le \gamma \le 1, \quad (5)$$

where β is a quantity that approaches 1.0 as α_g approaches 1.0, and also as the mixture velocity approaches a high value. We choose the velocity of the onset of the annular flow regime to be the velocity at which the profile slip vanishes. The transition to annular flow occurs when the gas superficial velocity V_{sg} reaches the "flooding" value V_{sgf} that is sufficient to prevent the liquid film from falling back against the gas flow. An expression for the flooding velocity is given below in Eq. 10. Accordingly, we choose the following expression for β:

$$\beta = \max\left(\alpha_g, F_v \frac{\alpha_g |V_m|}{V_{sgf}}\right). \quad (6)$$

The parameters A, B, and F_v can be tuned to fit the observations. A represents the value of the profile parameter in the bubble and slug flow regimes, and is originally set to be 1.2 in ECLIPSE.[3] B represents the value of the gas volume fraction, or the mixture velocity as a fraction of the flooding velocity, at which C_0 starts to drop below A. B is originally set to be 0.3. F_v is a multiplier on the flooding velocity fraction, originally set to be 1.0. Profile flattening can be made more or less sensitive to the velocity by adjusting the value of F_v.

Intuitively we would expect the gas superficial velocity $V_{sg} = \alpha_g V_g$ to increase with both α_g and V_m, so the relationship for C_0 must also satisfy

$$\frac{\partial}{\partial \alpha_g}(\alpha_g C_0) > 0 \quad \text{and} \quad \frac{\partial}{\partial V_m}(V_m C_0) > 0. \quad (7)$$

There may be problems with stability if this is not the case. Both of these criteria will be satisfied if $B<(2-A)/A$.

The Gas/Liquid Drift Velocity. We can derive an expression for the gas/liquid drift velocity by combining data on the limits of countercurrent flow under a variety of conditions and interpolating between them to avoid discontinuities. The method honors observations of gas/liquid relative velocities at low and high gas volume fractions, and it joins them with a "flooding curve" (Holmes[17]), as shown in **Fig. 2.** The relative velocity observations relate to the rise velocity of gas through a stationary liquid. From Eqs. 1 and 3 we can relate this to the drift velocity,

$$V_g(V_l = 0) = \frac{V_d}{1-\alpha_g C_0}. \quad (8)$$

At low values of α_g we use the rise velocity of a bubble through a stagnant liquid, which Harmathy[18] observed to be 1.53 V_c, where V_c is the characteristic velocity given by

$$V_c = \left(\frac{\sigma_{gl} g(\rho_l - \rho_g)}{\rho_l^2}\right)^{1/4} \quad (9)$$

where σ_{gl} is the gas/liquid interfacial tension.

At high values of α_g we use the "flooding velocity," defined as the gas velocity that is just sufficient to support a thin annular film

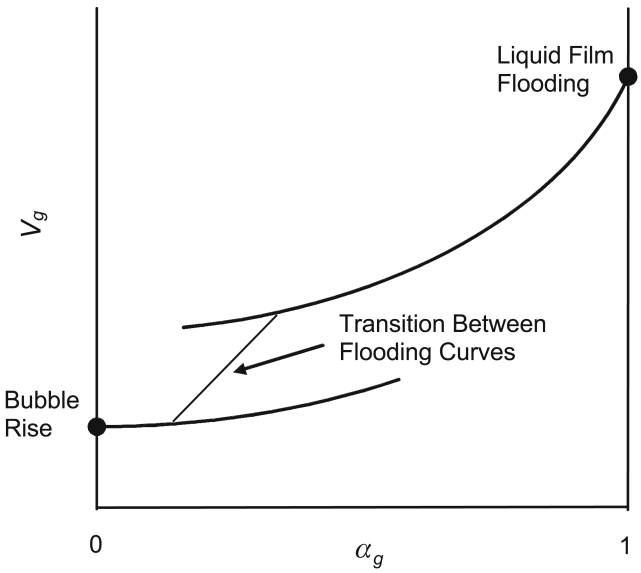

Fig. 2—Gas rise velocity in a stagnant liquid in the drift-flux model.

of liquid and prevent it from falling back against the gas flow. Wallis and Makkenchery[19] obtained the relation

$$V_g(V_l=0)=K_u\left(\frac{\rho_l}{\rho_g}\right)^{1/2}V_c, \quad (10)$$

where K_u is the "critical Kutateladze number," which is related to the dimensionless pipe diameter

$$\hat{D}=\left(\frac{g(\rho_l-\rho_g)}{\sigma_{gl}}\right)^{1/2}D, \quad (11)$$

according to **Fig. 3.**

To interpolate between these two extremes, we make use of the flooding curve described by Wallis[9] to define the limit of the countercurrent flow regime. Wallis observed that the gas and liquid flow rates that mark the limit of steady countercurrent flow lie on the curve

$$\left(\frac{\rho_g}{\rho_l}\right)^{1/4}\sqrt{V_{sg}}+\sqrt{-V_{sl}}=c\left[\frac{(\rho_l-\rho_g)}{\rho_l}gD\right]^{1/4}, \quad (12)$$

where c is a constant that depends on the pipe geometry. The sign convention is that upward flow is positive. We can normalize this curve to meet the flooding velocity observations in Eq. 10 as $\alpha_g \rightarrow 1.0$ by substituting the right side of Eq. 12 with $\sqrt{K_u V_c}$. Next, note that Eq. 1 can be written as

$$(1-\alpha_g C_0)V_{sg}-\alpha_g C_0 V_{sl}=\alpha_g V_d. \quad (13)$$

Assuming that C_0 does not vary with the flow velocity in the region of interest (the countercurrent flow region), for a given value of α_g Eq. 13 describes a straight line on a graph with axes V_{sg} and V_{sl}. Each of these lines should be tangential to the flooding curve—Eq. 12 renormalized—which represents the limit of countercurrent flow in the quadrant where $V_{sg}>0$ and $V_{sl}<0$. This requirement defines the drift velocity V_d as a function of α_g

$$V_d=\frac{(1-\alpha_g C_0)C_0 K_u V_c}{\alpha_g C_0\sqrt{\frac{\rho_g}{\rho_l}}+1-\alpha_g C_0}. \quad (14)$$

The curve must be "ramped" down in order to match Harmathy's bubble-rise velocity at low values of α_g. We apply a linear ramp between two selected values of the gas volume fraction a_1 and a_2. The overall relation for the drift velocity is thus

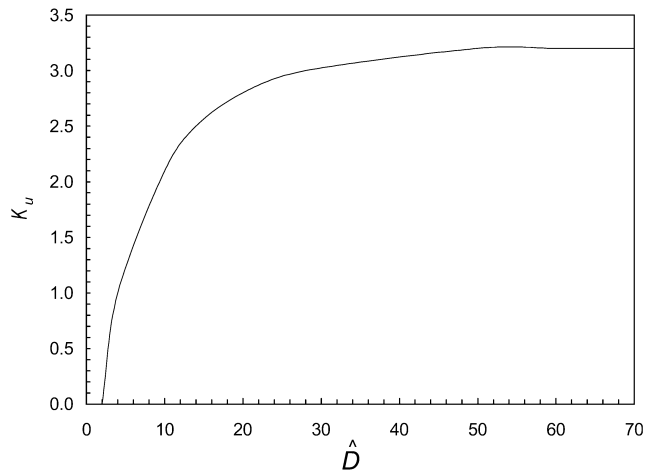

Fig. 3—Critical Kutateladze number as a function of dimensionless pipe diameter.

$$V_d=\frac{(1-\alpha_g C_0)C_0 K(\alpha_g)V_c}{\alpha_g C_0\sqrt{\frac{\rho_g}{\rho_l}}+1-\alpha_g C_0}, \quad (15)$$

where $K(\alpha_g)=1.53/C_0$ when $\alpha_g \leq a_1$, and $K(\alpha_g)=K_u(\hat{D})$ when $\alpha_g \geq a_2$, and a linear interpolation between these values when $a_1<\alpha_g<a_2$. Setting the values of a_1 and a_2 to 0.2 and 0.4, respectively, matches the data used by Zuber and Findlay[6] to demonstrate the transition from the bubble-flow regime.

Oil/Water Slip. The model described above was developed for gas/liquid flow in vertical pipes. Hasan and Kabir[12] proposed a drift-flux model for oil/water flow. The slip between oil and water is similarly described as a combination of profile and buoyancy effects:

$$V_o=C_0' V_l+V_d'. \quad (16)$$

For the oil/water profile parameter C_0', they suggest a value of 1.2, but this should decrease to 1.0 when oil becomes the continuous phase ($\alpha_o>0.7$). We make C_0' a continuous function of the oil volume fraction,

$$C_0'=A' \quad \text{when} \quad \alpha_o \leq B_1'$$
$$C_0'=1.0 \quad \text{when} \quad \alpha_o \geq B_2'$$
$$C_0'=A'-(A'-1)\left(\frac{\alpha_o-B_1'}{B_2'-B_1'}\right) \quad \text{when} \quad B_1'<\alpha_o<B_2'. \quad (17)$$

The parameters A', B_1', and B_2' are adjustable and are originally set to 1.2, 0.4, and 0.7 respectively. A condition equivalent to Eq. 7—that the oil superficial velocity should increase with the oil volume fraction—requires that $B_1'<(2-A')\,B_2'$.

For the oil-in-liquid drift velocity, they suggest

$$V_d'=1.53V_c'(1-\alpha_o)^2, \quad (18)$$

where V_c' is the characteristic velocity (Eq. 9) derived with oil and water properties,

$$V_c'=\left(\frac{\sigma_{ow}g(\rho_w-\rho_o)}{\rho_w^2}\right)^{1/4}. \quad (19)$$

For oil-dominated flow ($\alpha_o>0.7$), they recommend switching to a no-slip model. But because Eq. 18 has already reduced the drift velocity by an order of magnitude when the oil volume fraction reaches 0.7, we retain that relation over the full range of α_o in order to maintain continuity.

To model three-phase (oil, water, and gas) flow, our current treatment entails a two-stage approach that uses the available two-phase flow models. First we combine the oil and water into a single liquid phase with flow-weighted average properties and determine the flow velocities of the gas and liquid phases using Eq. 1. We then determine the oil and water velocities within the liquid phase

using Eq. 16. For this calculation, we use the oil volume fraction in the liquid phase, $\alpha_{ol}=\alpha_o/(\alpha_o+\alpha_w)$. This very simplistic approach ignores any effect that the presence of a third phase may have on the respective two-phase flow models, and so must be regarded as highly tentative. Nevertheless, it does produce the expected qualitative behavior, enabling a stagnant three-phase mixture to separate into gas, oil, and water zones through countercurrent flow. In recent work, we determined optimized drift-flux parameters for three-phase flow using the experimental data discussed in this paper. The three-phase model is reported in a separate paper[20] and will not be discussed here.

Inclined Flow. The relationships described previously are based on observations for vertical flow. Hasan and Kabir[12] proposed the following scaling for the terminal rise velocity for oil-water flow in inclined pipes:

$$V_{d\theta}=V_{d\theta=0}(\cos\theta)^{0.5}(1+\sin\theta)^2, \quad (20)$$

where θ is the angle of deviation from the vertical. The terminal rise velocity of a bubble or droplet is equivalent to V_d as α_o approaches zero. The rise velocity increases to approximately 2.5 times its value in vertical pipes, before falling off steeply to zero when the pipe is horizontal. Hasan and Kabir[12] presented their scaling as valid for oil/water flow for deviations $\theta<=70°$. The application of Eq. 20 outside these conditions and over the complete range of α can only be viewed as a tentative procedure.

Experimental Setup and Results

We now briefly describe the experimental setup and data that will be used for our determination of the drift-flux parameters. For a full description, see Oddie *et al.*[5]

Experimental Setup. The experiments were conducted in a 10.9-m-long, 15.2-cm-diameter, inclinable flow loop. The test section, shown in **Fig. 4,** is made of plexiglass to enable visual observations. During the experiments, the pipe deviation varied from 0° (upward vertical) to 92° (slightly downhill). Oil (kerosene with a viscosity of 1.5 cp at 18°C and a density of 810 kg/m^3) and water (tap water) were stored in a large separator and were transported separately to the inlet chamber. Another tank containing liquid nitrogen supplied gas to the pipe. This gas was first passed through evaporators and heat exchangers to allow it to reach ambient temperature. Oil, water, and gas entered the pipe and flowed along the test section, where the flow pattern was observed and the pressure and holdup (as described below) were measured.

Once steady state was achieved and the steady-state measurements were completed, the test section was shut in using fast closing valves. After the fluids settled, the flow loop was rotated to the vertical if necessary so that the final positions of the fluid interfaces could be measured directly from markings on the test section. This provided the shut-in holdup. Over most of the range in holdup, the error of this absolute volume measurement is less than 1%. However, volume fractions greater than 94% could not be measured accurately because the part of the test section near the outlet is manufactured from steel and so is not transparent.

The other two methods used to estimate the steady-state holdup were nuclear and electric probe measurements. The nuclear gamma densitometer, instrumented 7.5 m from the inlet (see Fig. 4), measures the mean density of the fluid, from which the steady-state holdup can be calculated. Ten electric probes were installed at various axial positions along the pipe. These probes detect the local fluid conductance, which can be used to determine the water holdup. The probes provide estimates for water holdup for both steady-state and transient flows (the period following shut-in constitutes the transient stage).

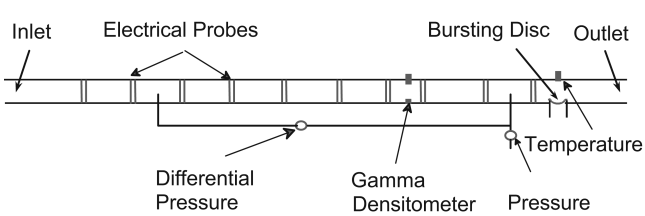

Fig. 4—Schematic of the test section of the flow loop.

The experiments were carried out over a wide range of flow rates for water, oil, and gas (designated Q_w, Q_o, Q_g). For water/gas flow, $2\leq Q_w\leq 100$ m^3/h and $5\leq Q_g\leq 100$ m^3/h was examined, while for oil/water flow $2\leq Q_o\leq 40$ m^3/h and $2\leq Q_w\leq 130$ m^3/h was investigated. For three-phase flows, the flow rate ranges were $2\leq Q_o\leq 40$ m^3/h, $5\leq Q_w\leq 40$ m^3/h, and $5\leq Q_g\leq 50$ m^3/h. Different combinations of flow rates were used for the different types of tests. Note that 1 m^3/h≈151 B/D; thus, the maximum-rate experiments correspond to flow rates of approximately 20,000 B/D. Each combination of flow rates was repeated at eight pipe deviations (0°, 5°, 45°, 70°, 80°, 88°, 90°, and 92° from upward vertical).

In our experiments, the entire test section is approximately 72 pipe diameters in length. For single-phase flow systems, fully developed flow is achieved at a distance of 50 to 100 pipe diameters from the inlet at a Reynolds number of 2500.[21] It is therefore not immediately clear whether our data can provide in-situ volume fractions for fully developed flow. For multiphase flow systems, however, it is unclear whether fully developed flow requires more or less distance (in terms of length to diameter ratio) to become established. It is possible that, at high Reynolds number ($\geq 10^4$), fully developed flow is achieved over less distance due to the turbulent interaction of the different phases. In any event, because our test section includes the inlet, at least some portion of the flow is not fully developed, even at high Reynolds number.

We do not believe, however, that this introduces major difficulties into our modeling work. In the first place, multiphase flow in wells (in the reservoir region) will also not be fully developed due to inflow effects (i.e., production from the reservoir into the well). Second, although the flow is not fully developed over at least a portion of the test section, the results for in-situ volume fraction appear to be fairly insensitive to this. Specifically, when we compared the flow pattern and holdup data with predictions from the Petalas and Aziz[15] mechanistic model for stabilized flow, we observed reasonably close agreement.[5] Furthermore, and most compellingly, for the experimental data used in this work, the three measurements of holdup show very good consistency, indicating that the effect of flow development is relatively small in terms of its impact on the holdup results. This issue will be discussed in detail in the next section.

Steady-State Holdup Results. Because holdup is one of the most important quantities characterizing multiphase flow in wellbores, three techniques (as discussed in the previous section) were used to assess steady-state holdup. We now compare the three measurements. In this study, two-phase steady-state holdup data for water/gas and oil/water flows are used for the determination of drift-flux parameters. We therefore present sample data only for these two types of flows.

We first demonstrate the relatively small impact of flow development on the holdup data. **Fig. 5** shows comparisons between shut-in holdup and probe holdup for an inclined oil/water system (upper curve and line) and a vertical water/gas system (lower curve and line). The curves display the time-averaged probe holdup data (where the averaging is performed only over the steady-state period) and the corresponding straight lines represent the shut-in holdup. An outlet effect (probe 10) is clearly present, and there is some variation in the probe data. Nonetheless, the general level of agreement between the probe and shut-in holdup data is quite close. If the effects of nonfully developed flow were significant, we would expect to see large variations in the probe holdup as a function of axial position along the pipe. By contrast, these results demonstrate that the shut-in holdup is in fact representative of the holdup along most of the pipe (even at the inlet) and that there are not large variations in holdup due to nonfully developed flow. A detailed consideration of the outlet effect (presented in Oddie *et al.*[5]) demonstrated that it introduces approximately a 5% rel-

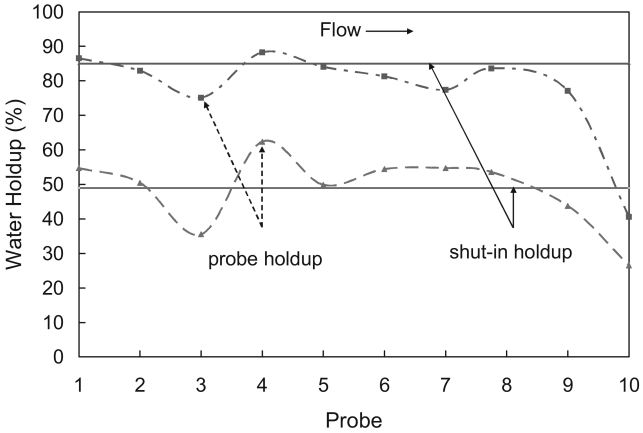

Fig. 5—Comparison of shut-in and probe water holdup (upper curve for θ=45°, Q_w=40.5 m^3/h, Q_o=10.0 m^3/h, H_w=85%; lower curve for θ=0°, Q_w=2.0 m^3/h, Q_g=60.2 m^3/h, H_w=49%).

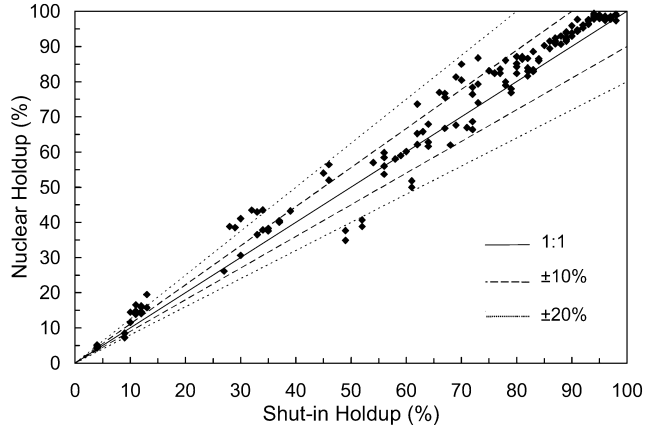

Fig. 6—Comparison of shut-in and nuclear holdup for water/gas system.

ative error in the shut-in holdup for two-phase flows over the range $0 \leq \theta \leq 90°$.

The lack of variation of holdup with position in the pipe is further demonstrated by comparing the shut-in and nuclear holdup results. The data for all inclinations for water/gas systems are presented in **Fig. 6.** The nuclear holdup values are computed by time-averaging the gamma densitometer data. Recall that the gamma densitometer is installed 50 pipe diameters from the inlet and that the nuclear measurements are very accurate, though the data must be interpreted (i.e., viewed as homogeneous, stratified, or intermittent) to determine the holdup values (see Oddie *et al.*[5] for detailed discussion). The nuclear holdup values in Fig. 6 are computed based on a stratified flow assumption, which was found to provide the best accuracy (even though the flows are of course not all stratified). The overall agreement between the nuclear and shut-in holdup results is reasonably good—91% of the data fall within the ±20% range. Viewing the flow as homogeneous or intermittent introduces relatively minor differences for most flows and does not alter the basic conclusions (see Figs. 12 through 14 in Oddie *et al.*[5]). From Figs. 5 and 6, we can reasonably conclude that the holdup does not vary excessively over the pipe length (with the exception of an outlet effect due to the pipe instrumentation) and that the effects of flow development on the holdup results are relatively small.

Another potential concern with the shut-in holdup measurements is the possible inaccuracy incurred by capturing an incomplete slug. This would pose a problem if the slugs were of length comparable to that of the test section. Typical experimental slug lengths were less than 2 m, which is relatively small compared to the 10.9-m test section length. The longest slugs encountered were approximately 2.3 m in length, which is only approximately 21% of the test section length. In addition, we did not observe large discrepancies in comparisons between shut-in holdup and the nuclear measurements for systems with large slugs (i.e., there is not any apparent correlation between the scatter in Fig. 6 and large slug lengths). Thus, we conclude that the holdup data are not discernibly impacted by the effects of incomplete slug capture.

Because the shut-in measurement is a direct measurement, with a measurement error of less than 1% for holdup values less than 94%, we used it to represent steady-state holdup. We now present a few experimental results for shut-in water holdup (referred to simply as "holdup") to provide an indication of the type and range of experimental data that will be used to determine the drift-flux model parameters.

Fig. 7 displays experimental results for water/gas systems. We present results at vertical and 45° deviation for three gas flow rates. This and subsequent plots show data in terms of water holdup vs. the input water fraction or water cut (c_w). For a water-gas system, $c_w = Q_w/(Q_w+Q_g)$. The error bars indicated in the figures are ±5%, as determined from repeated experiments. The vertical distance from any point to the $\alpha_w = c_w$ line gives the slip between the two phases. As gas flow rate increases, slip decreases for constant input fractions. The slip also decreases as c_w increases and as the pipe deviation shifts from 45° to vertical. As shown in **Fig. 8,** when the pipe is further deviated from 45° to 80°, the slip decreases. This, taken along with our measurements at 70° deviation (Oddie *et al.*[5]), indicates that the slip displays a maximum between 45° and 70°. As we will see, the drift-flux parameters determined from the data capture this effect.

Experimental results for oil/water systems, for vertical flow and flow at 70° deviation, are presented in **Fig. 9.** Qualitatively similar

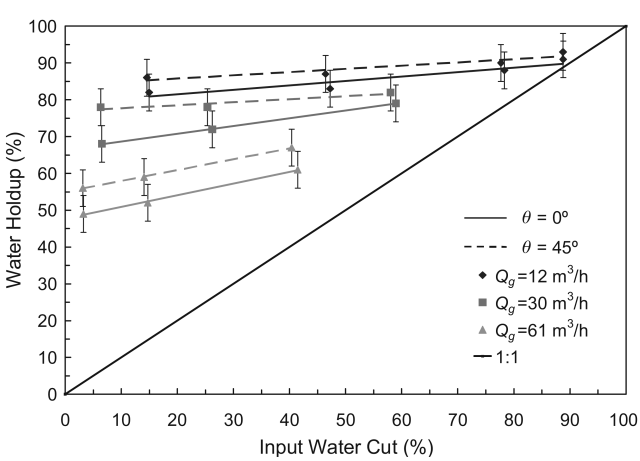

Fig. 7—Holdup for water/gas system for θ=0° and θ=45°.

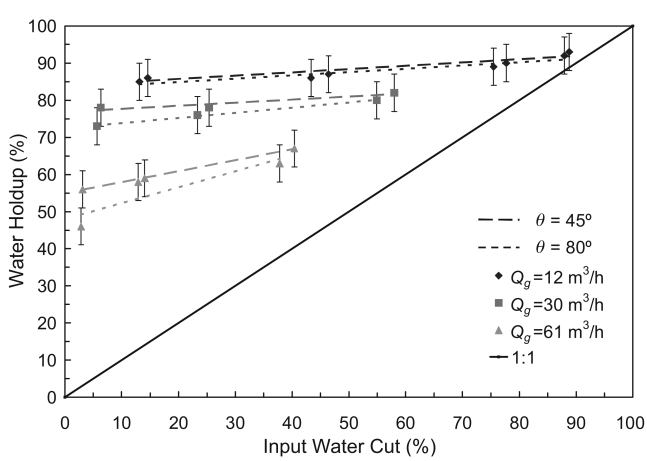

Fig. 8—Holdup for water/gas system for θ=45° and θ=80°.

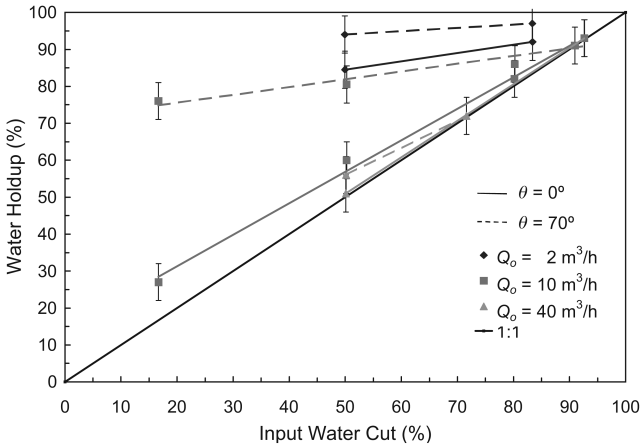

Fig. 9—Holdup for oil/water system for θ=0° and θ=70°.

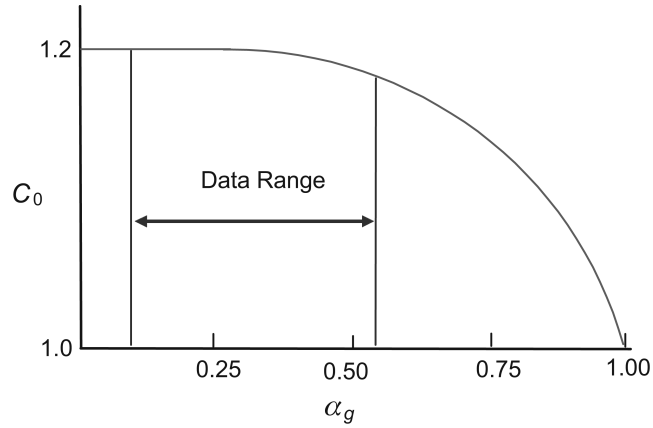

Fig. 10—Default profile parameter for water/gas system.

trends can be observed here as in Figs. 7 and 8, though the magnitude of the effects is quite different. For example, at high flow rates ($Q_o = 40$ m^3/h), there is very little slip at either inclination. At intermediate flow rates ($Q_o = 10$ m^3/h), by contrast, the effect of inclination is very pronounced, with much higher slip observed at 70°. Again, our intent is to capture these complex effects through the optimized drift-flux parameters.

Parameter Determination

In this section, we describe the optimization procedure used to determine the drift-flux parameters. The parameters are determined such that the difference between the experimental data and model predictions for holdup are minimized in a least-square sense. We first discuss water/gas systems and then consider oil/water flows. For both systems we use experimental data for only the six deviations from 0° to 88° (inclusive) for the parameter determination. This is because the experimental holdup data for 90° and 92° display relatively large errors because of the way in which fluid exited the test section (Oddie *et al.*[5]).

From our earlier description of the drift-flux model, it is apparent that there are many parameters that could potentially be modified to generate close fits between the experimental and model predictions. However, we wish to avoid overly tuning the model because this may result in some parameters assuming unphysical values or in an unnecessarily complex description. We therefore performed a number of optimizations holding different sets of parameters fixed (and varying others) to determine an appropriate balance between model complexity and closeness of fit.

The parameters we ultimately chose to optimize were selected primarily based on two considerations. We optimized parameters to which the model is most sensitive as well as parameters that are the most uncertain. For example, for the water/gas system, seven parameters were optimized. These include A and B, which define the shape of the C_o vs. α_g curve (see **Fig. 10**), as well as a_1 and a_2, which define the transitions in the V_d vs. α_g curve (**Fig. 11,** generated using data for a 15-cm-diameter pipe). The remaining parameters optimized define the drift velocity multiplier m. This multiplier is especially useful in matching data from inclined pipes, where the original relationship is given by Eq. 20. Specifically, from Eq. 20, the default representation for $V_{d\theta}$ can be written as

$$V_{d\theta} = V_{d\theta=0} m(\theta), \quad (21)$$

where $m(\theta)$ gives the inclination effect

$$m(\theta) = (\cos\theta)^{0.5}(1 + \sin\theta)^2. \quad (22)$$

We parameterize $m(\theta)$ as

$$m(\theta) = m_0(\cos\theta)^{n_1}(1 + \sin\theta)^{n_2}. \quad (23)$$

Here, m_0 is used to account for a nonunit multiplier for vertical flow; m_0, n_1, and n_2 are all adjustable parameters.

The model is sensitive to all seven of these parameters. The a_1 and a_2 transition parameters are quite uncertain, because they were introduced to interpolate between models describing the two limiting cases.

The vector X_P containing the seven parameters to be optimized for the water/gas system is as follows:

$$X_P = [A \quad B \quad a_1 \quad a_2 \quad m_0 \quad n_1 \quad n_2]^T$$

The parameters are tuned to minimize the error between the experimental α_g and the α_g estimated from the drift-flux model (designated α_g^*) for water/gas flows. The objective function is

$$E_{\alpha_g} = \sum_{i=1}^{N} w_i |\alpha_{gi} - \alpha_{gi}^*|^2, \quad (24)$$

where N is the total number of experimental data points and w_i is the weight assigned to each point (i.e., $w_i = 0.5$ for each of two repeated tests) and the experimental error. For example, experimental α_g (or α_o) less than 6% are assigned a very small value of w_i because the exact value could not be measured in this range. The minimization of E_{α_g} in Eq. 24 is a nonlinear least-square problem. We apply a trust region reflective Newton algorithm in MATLAB[22] for this minimization.

The calculated in-situ gas volume fraction, computed by

$$\alpha_{gi}^* = \frac{V_{sgi}}{C_o(\alpha_g^*, X_P)V_{mi} + V_d(\alpha_g^*, X_P)} \quad (25)$$

is obtained from the drift-flux model and the vector of parameters X_P. Because Eq. 25 is implicit in α_{gi}^*, it must be solved using an iterative procedure. In this work, a successive substitution technique is applied. The initial guess for α_{gi}^* in Eq. 25 is the experimental value α_{gi}^*. The ranges over which the parameters in X_P may change during the optimization are specified to limit them to

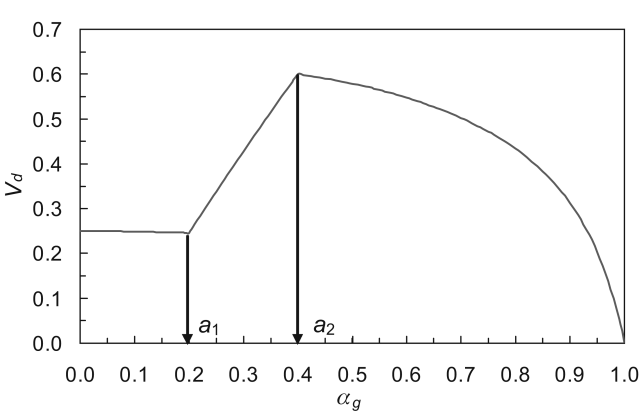

Fig. 11—Default drift velocity for water/gas system in 15.2-cm diameter pipe.

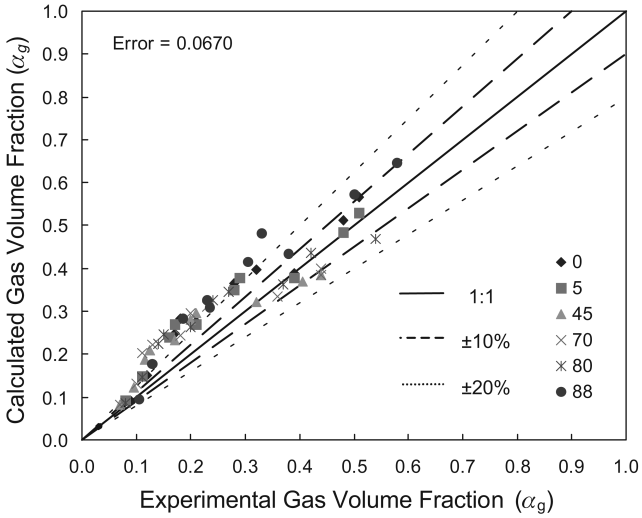

Fig. 12—Predicted in-situ gas volume fraction using original parameters (A=1.2, B=0.3, a_1=0.20, a_2=0.40, Eq. 22).

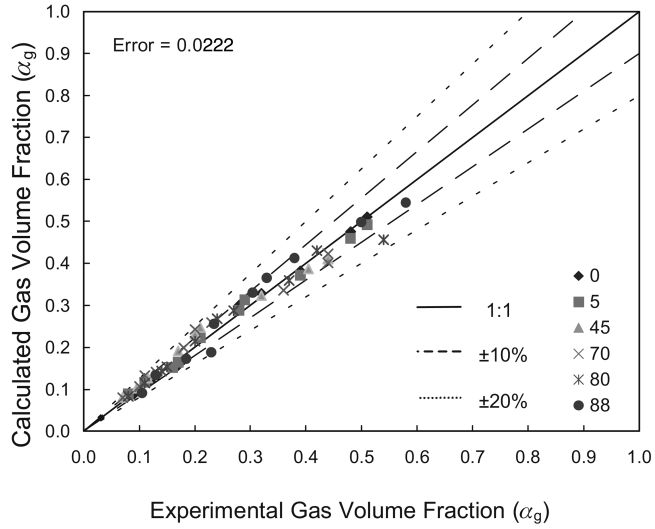

Fig. 13—Predicted in-situ gas volume fraction using optimized parameters (A=1.0, a_1=0.06, a_2=0.21, Eq. 26).

physically allowable values. For example, we require $1.0 \leq A \leq 1.5$ to maintain consistency with previous findings.

For oil/water systems, we again select seven parameters for optimization. Three of these parameters are analogous to those considered above: A', B'_1, and B'_2 (which define C'_0). We introduce an additional parameter in the expression for the oil/water drift velocity. This new parameter, designated n', replaces the exponent 2 in Eq. 18. The drift-flux multiplier $m'(\theta)$ is parameterized as

$$m'(\theta) = n'_1 \cos\theta + n'_2 \sin 2\theta + n'_3 \sin 3\theta, \quad \text{(26)}$$

where n'_1, n'_2, and n'_3 are adjustable parameters. The vector of parameters to be optimized, X'_P, then is:

$$X'_P = [A' \quad B'_1 \quad B'_2 \quad n' \quad n'_1 \quad n'_2 \quad n'_3]^T.$$

For both water/gas and oil/water flows, we determine the optimal parameters using the combined data from all six deviations. Because the optimization results may depend on the starting values, the optimization is run many times (typically more than 1,000) with different initial estimates for the parameters. Using this approach, we can search the parameter space more completely. The technique does not, however, guarantee that a global minimum will be reached.

We note finally that the drift-flux parameters determined in this work are expected to be applicable for pipes with diameters close to that considered here (in the range of 15.2 cm). This size is of practical relevance for reservoir operations. These parameters may not be appropriate for modeling flow in wells of different diameters. In addition, these parameters will likely require some modification for fluid systems that are very different from that considered here.

Results and Discussion

We now present the results for the parameter optimizations for water/gas and oil/water flows. We present results in terms of calculated α_g or α_o vs. experimental α_g or α_o. Error is quantified as the root mean square average of the absolute error of each point.

Water/Gas Flows. As discussed in the Introduction, the original parameters currently used in the ECLIPSE simulator were estimated, in many cases, from experiments on small-diameter (e.g., 2-in.) pipes. Thus, we would expect there to be some degree of error when these parameters are applied for large-diameter pipeflow. The predictions for α_g for deviations over the range $0 \leq \theta \leq 88°$, using these original parameters, are shown in **Fig. 12.** Note that we only include data for α_g>0.06. Also shown in the figure are dashed lines designating errors of ±10% and ±20%. There is a clear tendency toward overprediction for $0.1 < \alpha_g < 0.3$. The error is 0.067.

Using the optimized parameters, we achieve the results shown in **Fig. 13.** The error here is reduced considerably to 0.0222, and all of the data now fall within the ±20% range. This improvement appears to be largely due to the reduction of a_1 and a_2 from their default values. It is worth reiterating that only V_d depends on the pipe inclination (through m) and that none of the parameters related to C_0 varies with θ.

The optimized parameters used for these predictions are:

$$A = 1.0,\ a_1 = 0.06,\ a_2 = 0.21, \text{ and}$$

$$m(\theta) = 1.85(\cos\theta)^{0.21}(1 + \sin\theta)^{0.95}. \quad \text{(27)}$$

Note, in contrast to $A = 1.2$, as appropriate for small-diameter systems, the best fit here is $A = 1.0$, which means that $C_0 = 1$ for the entire range of α_g, and the value of B does not affect the model. This indicates that there is no profile effect. As we discussed before, it is possible that our results do not show profile slip because the flow is not fully developed.

However, if we forced $A = 1.2$ and $B = 0.60$, we obtain

$$a_1 = 0.06,\ a_2 = 0.12,$$

$$m(\theta) = 1.27(\cos\theta)^{0.24}(1 + \sin\theta)^{1.08}. \quad \text{(28)}$$

The error with these parameters is 0.0264 (compared to 0.0222 in Fig. 13). This illustrates that different sets of parameter values can provide fits of similar accuracy. Therefore, whether $C_0 = 1$ or not does not appear to be a major concern here, because the predicted α_g are in close agreement with the experimental data in either case.

The variation of m with θ is shown in **Fig. 14.** Eq. 27, obtained from the optimization, is shown as the solid curve. Note that Eq. 27 is valid up to 88° and should not be taken to imply that V_d is zero for horizontal flow. Observations of nonzero V_d have been made for small-diameter systems.[23] The $m(\theta)$ suggested by Hasan and Kabir[12] for oil/water flow (Eq. 22) is also shown in Fig. 14 as the dotted curve. As indicated by comparing the two lines, the optimized model and the original model in ECLIPSE, V_d is much higher in the optimized model. This is mostly because the value of the corresponding C_0 is 1.0 for Eq. 27. When we force $A = 1.2$, Eq. 28 gives a result for $m(\theta)$ similar to that provided by the Hasan and Kabir[12] correlation. This result is shown in Fig. 14 as the dashed curve.

Oil/Water Flows. We next consider oil/water systems. The predictions for α_o using the original parameter values are shown in **Fig. 15.** The error of 0.135 here is quite significant. A cluster of four points (corresponding to calculated α_o of ~0.75) is noticeably outside the 20% error line. These points all correspond to flows in which $Q_o = 10$ m^3/h and $Q_w = 2$ m^3/h. These flow rates are relatively low, but the input oil cut is the highest investigated. The

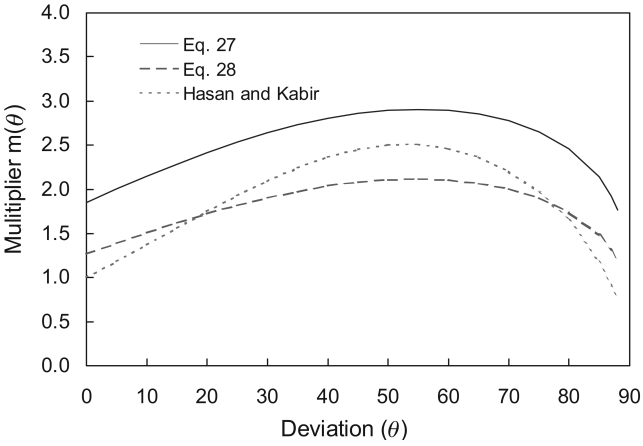

Fig. 14—Deviation effect for water/gas system.

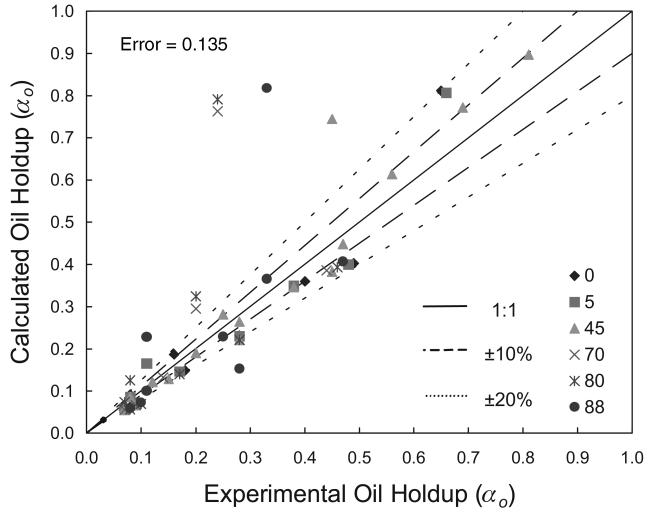

Fig. 15—Predicted oil holdup using original parameters (A'=1.2, B_1=0.4, B_2=0.7, n'=2.0, Eq. 22).

original parameter values result in significant errors for this combination of flow rates because V_d is significantly underestimated, as we will see here.

Predictions using the optimized parameters are shown in **Fig. 16.** Using the optimized set of parameters,

$A' = 1.0$, $n' = 1.0$, and

$$m'(\theta) = 1.07 \cos\theta + 3.23 \sin 2\theta - 2.32 \sin 3\theta, \quad (29)$$

the average error is reduced to only 23% of that using the original parameters, and the four outliers now cluster around the diagonal. A few points remain outside of the ±20% range, however. When $A' = 1.0$, the parameters B_1' and B_2' do not enter the model because C_0' remains constant at 1.0 (see Eq. 17). Our finding that $A' = 1.0$ (which implies that $C_0' = 1.0$ independent of α_o) is consistent with the earlier result of Hill.[24,25] He considered oil/water flow in a 16.4-cm-diameter vertical pipe (the oil was kerosene with a density of 780 kg/m^3 and a viscosity of 2 cp). Using linear regression, Hill found that $C_0' = 1.0$. Our result is in close agreement with this observation.

The variation of m' with θ (Eq. 29) is shown in **Fig. 17** as the solid curve. The original correlation (the dotted curve in the figure) significantly underestimates $m'(\theta)$ (and thus V_d) at $50° \leq \theta \leq 88°$. This underestimation explains some of the error in Fig. 15.

Table 1 below summarizes our recommended parameter values for two-phase water/gas and oil/water flows.

Conclusions

The drift-flux model is well suited for use in reservoir simulators because it is relatively simple and because it provides a continuous and differentiable representation of multiphase flow in pipes and wells. In this work, we used recently collected experimental data for large-diameter pipes to determine drift-flux parameters for wellbores and pipes. The following conclusions can be drawn from this study:

- Drift-flux parameters determined from experiments using small-diameter pipes (2 in. or less) may not be appropriate for use in models of flow in large-diameter wellbores.
- Drift-flux descriptions with optimized parameters can accurately describe flow in large-diameter vertical and inclined pipes for both water/gas and oil/water systems. Optimized parameters can be obtained using a unified optimization approach, which yields parameters applicable for the entire data range, including all six deviations.

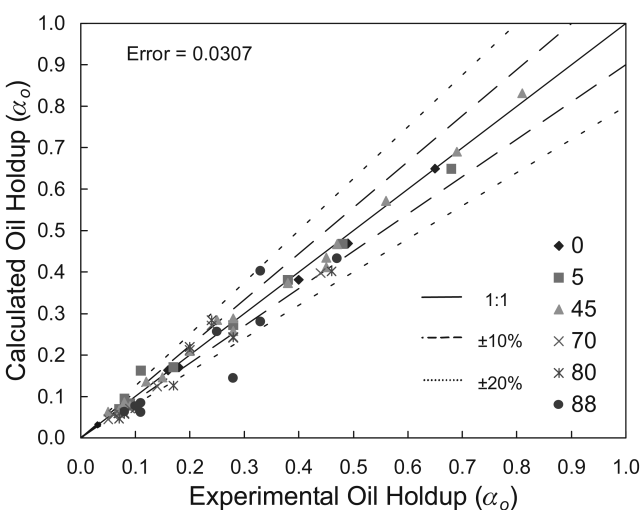

Fig. 16—Predicted oil holdup using optimized parameters (A'=1.0, n'=1.0, Eq. 29).

Nomenclature

a_1 = drift velocity ramping parameter
a_2 = drift velocity ramping parameter
A = profile parameter for low gas fraction in liquid
A' = profile parameter for low oil fraction in water
B = profile parameter term, gas volume fraction at which C_0 begins to reduce
B_1 = profile parameter term, oil volume fraction at which C_0' begins to reduce
B_2 = profile parameter term, oil volume fraction at which C_0' falls to 1.0
c = input volume fraction

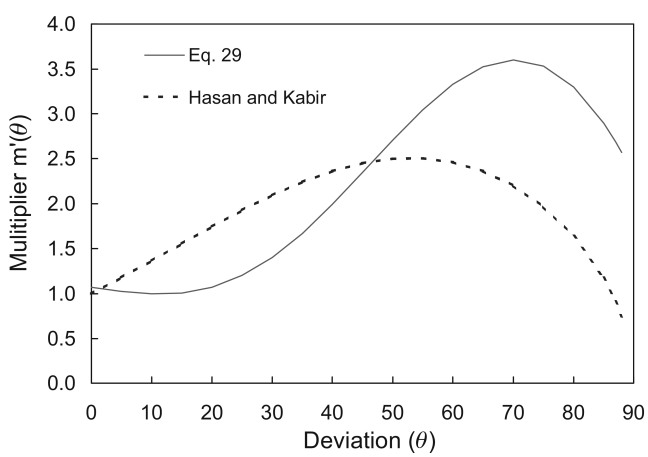

Fig. 17—Deviation effect for oil/water system.

TABLE 1—SUMMARY OF RECOMMENDED PARAMETERS FOR WATER/GAS AND OIL/WATER SYSTEMS

Water/Gas				
A	B	a_1	a_2	$m(\theta)$
1.00	—	0.06	0.21	$1.85(\cos\theta)^{0.21}(1+\sin\theta)^{0.95}$
Oil/Water				
A'	B_1	B_2	n	$m'(\theta)$
1.00	—	—	1.00	$1.07\cos\theta+3.23\sin2\theta-2.32\sin3\theta$

C_0 = profile parameter for gas/liquid flow
C_0' = profile parameter for oil/water flow
D = pipe internal diameter
F_v = velocity sensitivity parameter for liquid/gas flows
g = gravitational acceleration
K_u = critical Kutateladze number
m = drift velocity multiplier for water/gas flow
m' = drift velocity multiplier for oil/water flow
n = Hasan and Kabir deviation effect parameter
n' = Hasan and Kabir drift velocity exponent
N = number of experimental points
Q = flow rate
V = velocity
V_c = characteristic velocity
V_d = gas drift velocity in gas/liquid flow
V_d' = oil drift velocity in oil/water flow
V_m = mixture velocity
V_s = superficial velocity
X_P = vector of parameters optimized
α = in-situ volume fraction
β = profile parameter reduction term
γ = profile parameter reduction term
θ = deviation from vertical
ρ = density
σ = interfacial tension/surface tension

Subscripts

g = gas
l = liquid
m = mixture
o = oil
w = water

Acknowledgments

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SI Metric Conversion Factors

cp × 1.0*	E–03	= Pa·s
ft × 3.048*	E–01	= m
°F (°F – 32)/1.8		= °C

* Conversion factor is exact.

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